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Tasks

1. Mean aggregate spike rate:

$$R = 1024 \text{ nerons} \times 50 \frac{\text{spike}}{\text{s}} = \boxed{51200 \frac{\text{spike}}{\text{s}}}$$

2. $B = R \times 20 \frac{\text{bits}}{\text{spike}} = (51200 \text{ spikes/s}) \times 20 \frac{\text{bits}}{\text{spike}} = 1024000 \text{ bits/s}$

$= \boxed{1.024 \text{ Mbits/s}}$

3.

Interface	Range (Mbits/s)	Sustainable?
I^2C	≤ 3.4	Y
SPI	≤ 50	Y
AXI4-Lite	≤ 100	Y

The Lowest complexity interface that suffices the bit rate is I^2C.

4. $B_{Burst} = \dfrac{256 \text{ neurons} \times 1000 \frac{\text{spikes}}{\text{s}} \times 20 \frac{\text{bits}}{\text{spike}}}{10^6 \frac{\text{bits/s}}{\text{Mbits/s}}} = \boxed{5.12 \frac{\text{Mbits}}{\text{s}}}$

$\dfrac{B_{Burst}}{B} = \dfrac{5.12 \text{ Mbits/s}}{1.024 \text{ Mbits/s}} = \boxed{5\text{-to-}1}$

B_{Burst} exceeds the I^2C bit rate range and buffering of a few bytes deep is required.

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5. $B_{Frame} = 1024 \frac{bits}{spike} \times 1000 \frac{spikes}{s} = \boxed{1.024 \ Mbit/s}$

$$\frac{B_{Frame}}{B} = \frac{1024000 \ bits/s}{1024000 \ bits/s} = \boxed{1\text{-to-}1}$$

$$1024_{neurons} \times f_{crossover} \times 20 \frac{bits}{spike} = 1024_{neurons} \times f_{crossover} \times 1 \frac{bit}{neuron\text{-}spike}$$

$$f_{crossover} = 50 Hz$$

This implys that AER is better for Neural networks that have less event frequency.